

Role

Project lead

Ended up also creating, implementing, and animating UI along with creating visual effects for the project.

Challenges

Time. Alongside this class I was working about 12 hours per day on MQP (not by choice). This meant I had very little time and energy for this project despite enjoying working on it. I found it was difficult to find the time to work on this project outside of the scheduled class time.

Additionally, our version control provided some challenges as there was a period of time where we were experiencing non-stop merge conflicts that we were unable to deduce the cause and unable to solve.

How these were overcome

To overcome this I had to be efficient. I had to figure out how to get a lot done in very little time. Luckily, I know Unreal pretty well and was able to find the things I needed quickly, but in the case of tutorials I had to put them at 2x speed and program as quickly as I could.

For the version control, we had to just rebuild anything that was conflicted or lost one by one in main, making sure no one else fiddled with the same blueprint while someone else was rebuilding with it in main.

Lessons Learned

I believe the best lesson learned was one that my team did well. With the entire programming team composed of seniors, we leaned towards a functionally simple game with an emphasis on visuals such as particles, shaders, animations, lighting, etc. This decision ended up being a lifesaver because, aside from me, other programming members also had little time. Thus, less functionality meant the core game was done quickly and provided little stress.

Additionally, another lesson learned is not being final on our genre prior to development. My team decided to be flexible with our genre when talking to art teams. While a “magical girl” themed game isn’t typically associated with a “horror” genre, we had a ton of fun fitting in horror elements into our magical and cutesy game.

In the future, I believe students, especially seniors, should estimate and evaluate their workload and adjust the project’s goals accordingly and be open-minded to genres.

Version Control

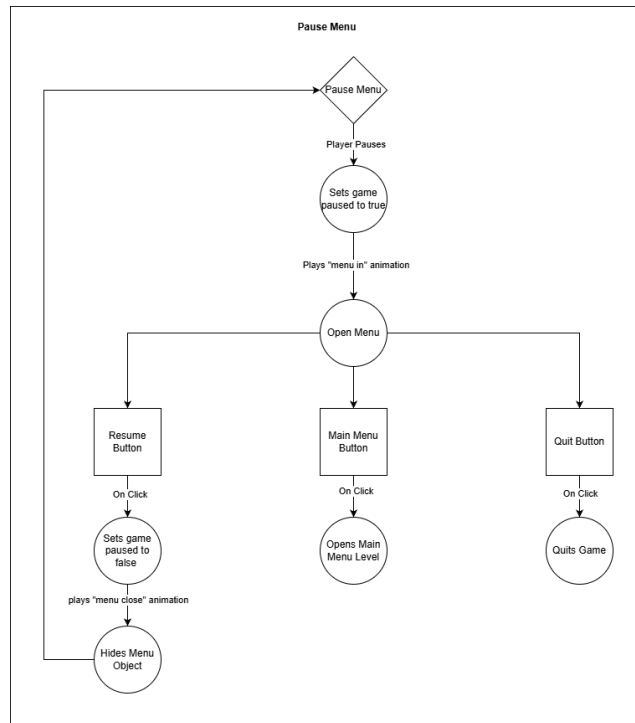
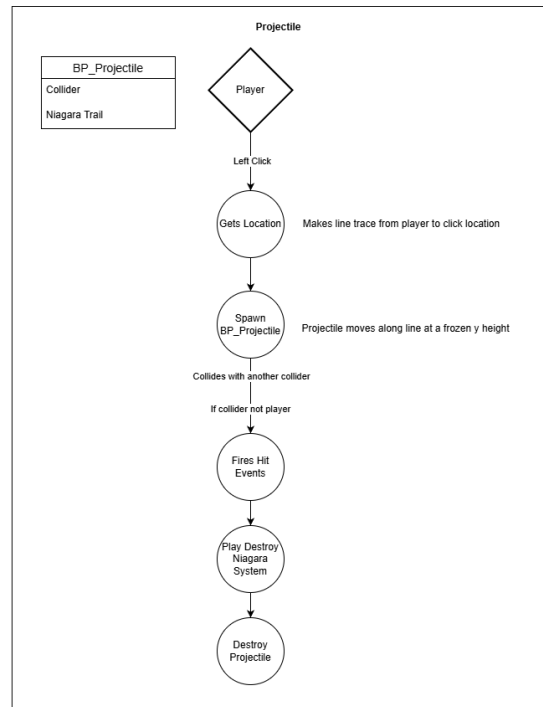
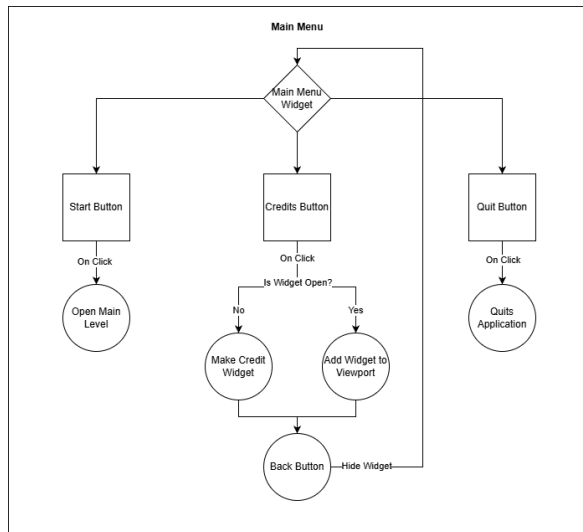
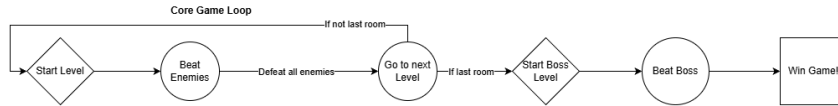
Initially, there was discussion about using Perforce as it is easier to use with Unreal Engine. However, there was a risk of setup taking more time so the group decided on using GitHub.

While the overall experience was overall stress free, there was a time-frame in the project where we were having non-stop merge conflicts with a level no one in the team touched.

Continuing with the merge would result in many functions being missing and thus needing to be rebuilt by the team. While we were eventually able to get around this issue, it was one of the largest roadblocks in the project.

Architectural Diagram

General Diagram & Lada Architecture



Anu Architecture

